



RICE

Yuheng Wang

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EDUCATION

- **Rice University** Aug. 2026 – Present
Ph.D. in Applied Physics Houston, USA
- **University of Michigan** Sep. 2024 - May. 2026
Undergraduate in Engineering Physics, minor in Computer Science Ann Arbor, USA
GPA: 3.96/4.00; Summa Cum Laude
- **Shanghai Jiao Tong University** Sep. 2022 - August 2026
Undergraduate in Electrical and Computer Engineering Shanghai, China
GPA: 3.77/4.00; Outstanding Graduate

RESEARCH INTEREST

I am broadly interested in the intersection of physics and computation. My primary focus lies in quantum physics, cavity QED and photonics, with an emphasis on understanding fundamental physical principles through simulation and modeling. Leveraging my dual background in Engineering Physics and Electrical & Computer Engineering, I am also interested in developing and applying artificial intelligence and machine learning methods to accelerate scientific discovery, such as enhancing physical simulations, analyzing complex experimental data, and bridging theoretical models with experimental observables.

RESEARCH EXPERIENCE

- **University of Michigan, Department of Radiation Oncology** May. 2025 - May. 2026
Undergraduate Research Assistants Ann Arbor, USA
 - Advisor: Lise Wei
 - Equating the Biologically Effective Dose of SBRT and 90Y SIRT
 - * Modeled subvoxel SIRT nonuniformity via EUD, then fit mixed-effects logistic dose-response with covariates; calibrated repair rate and micro-dose CV by aligning SBRT/SIRT curves with bootstrap uncertainty.
 - * Has been selected for presentation in a Quick Pitch Oral Scientific Session during the 2026 ASTRO's Annual Meeting: "Toward a Common Biological Dose Scale for SBRT and SIRT Using Imaging-Derived Liver Functional Toxicity Response".
- **University of Michigan, Deng Research Lab** Sep. 2024 - Aug 2025
Undergraduate Research Assistants Ann Arbor, USA
 - Advisor: Chulwon Lee, Faculty Instructor: Hui Deng
 - Loss-Induced Synthetic Magnetic Field
 - * Designed & simulated rotationally asymmetric Si ring-resonator arrays to engineer loss-induced synthetic magnetic fields; observed two-sided non-Hermitian skin-effect localization in 3D FDTD.
 - * Built a non-Hermitian tight-binding model; eigen-analysis identified zero net loop phase as the barrier to skin-effect; proposed phase/loss asymmetry and distance-offset remedies and validated designs in simulation.
 - * Tools: SYS Lumerical FDTD, MATLAB; GrateLake HPC.
 - Non-Hermitian Exciton-Polariton Transport beyond Lorentz Reciprocity Theorem
 - * Fall 2024 - Winter 2025 Undergraduate Research Opportunity Program
 - * Using Ansys Lumerical to simulate a nanobeam cavity integrated with a grating coupler
 - * Exfoliating MoSe2 to make monolayer sample

* Manuscripts submitted: Lee, C., Lee, J.H., Zhang, K., Wang, Y., **Wang, Y.**, Lydick, N., Miao, J., Xin, X., Liu, Y., Sun, K., & Deng, H. "Harnessing nanophotonic edgemodes by engineering PT symmetry"

- **Shanghai Jiao Tong University, Laboratory of Ultrafast Integrated Systems (LUIS)** Aug. 2023 - May 2024
Undergraduate Research Assistants Shanghai, China
 - Advisor: Xuyang Lu
 - Laboratory website: <https://sites.gc.sjtu.edu.cn/xuyang-lu/>
 - ENGR4903J Spring 2024: Chip-scale switch-mode wireless power transfer system based on parity-time method
 - * Use the Kuramoto model and Adler equation to design an array of oscillators that can change the mutual phase difference by adjusting the coupling coefficient
 - * Understand the basic usage of Cadence and the methods associated with analog circuits
 - ENGR4903J Fall 2023: Machine-learning method for mmWave based voice and vital signal recognition
 - * Implement the VMD-HHT algorithm to filter noise better and reconstruct the received signal
 - * Learn the underlying principles of microcontroller systems, how signals are sent and received between microcontrollers

TEACHING AND MENTORING EXPERIENCE

- **JI Center for Learning and Teaching, Shanghai Jiao Tong University** Sep. 2023 - Aug. 2024
ENGR151(Accelerated Introduction to Programming), ECE311(Analog Circuits) Teaching Assistant Shanghai, China
 - Hold RC to students as ENGR151 Teaching Assistant
 - Hold Lab to students as ECE311 Teaching Assistant
 - Contribute to the scripts for the teaching team to automate workflows

PROJECTS

- **Coupled Flow Matching (CPFM): Controllable Dimensionality Reduction & Reconstruction** May. 2025 - Oct. 2025
<https://arxiv.org/abs/2510.23015> Ann Arbor, USA
 - CPFM, a framework that learns coupled continuous flows for data x and low-dim embeddings y to enable both $p(y|x)$ sampling (latent-space flow) and $p(x|y)$ reconstruction (data-space flow).
 - Extended Gromov–Wasserstein OT with a kernelized quadratic objective to encode semantic priors; derived an alternating minimization with entropic Sinkhorn, achieving $O(n^2)$ per-iteration complexity with an ε -scheduling scheme for stability.
- **Analogical analyses of epidemic evolution from the fluid mechanics view** Sep. 2022 - May. 2024
The 26th ‘Shanghai Jiaotong University Student Innovation and Practice Programme’ Project Shanghai, China
 - Model epidemic evolution using meta cellular automate written in Python and Matlab
 - Study second-order phase transitions and link to outbreaks of epidemics

TECHNICAL SKILLS

Languages: Lumerical, MATLAB, PyTorch, HPC, scientific computing

Miscellaneous: L^AT_EX, Cadence, Verilog, Russian

ACHIEVEMENTS

2026 **James B. Angell Scholar**, University of Michigan
2025 **Michigan Engineering Continuing Student Scholarship**, Roger King Scholarship
2024–2025 **University Honors**, University of Michigan
2024–2025 **Dean’s List**, University of Michigan
2024 **SJTU Undergraduate Excellent Scholarship**, Academic Year 2023–2024
2024 **Meritorious Winner**, COMAP Mathematical Contest in Modeling (MCM)
2023 **SJTU Undergraduate Excellent Scholarship**, Academic Year 2022–2023
2020 **First Prize**, Chinese Physics Olympiad (CPhO), Guangdong Province